



Shri Guru Gobind Singhji Institute of Engg. & Technology, Vishnupuri, Nanded-431606.
Department of Production Engineering.
In-Semester Examination / Evaluation I
Academic Year: 2025-26 (semester I)

Class: Final Year B.Tech. Course: CAD/CAM & Robotics. Course Code: PEC-PE404

Date: 19.08.2025 Time: 11:00 to 11:45 Max. Marks: 20

Note:

Reg.No:

- i. Assume suitable data whenever necessary and state it clearly.
- ii. Mobile phones and programmable calculators are strictly prohibited.
- iii. Exchange/Sharing of stationery, calculator etc. not allowed.
- iv. Figures to right indicate full marks.
- v. The symbols carry usual meaning.

Q.N.			Marks	CO	BT
Q.1	a	Explain the terms: Motherboard, Microprocessor, RAM and ROM.	04	CO1	BT1
	b	Explain CAD, CAM and CAE & their packages of each.	06	CO2	BT2
Q.2	a	Derive 2D Rotation transformation.	04	CO2	BT3
	b	A polygon through vertices A (0,0), B (1,0), C (1,1), D (0,1). Perform the following transformation and conclude the result of shearing: a) x-shearing with $Sh_x=2$ followed y-shearing with $Sh_y=3$. b) Simultaneous x and y shearing with $Sh_x=2$ and $Sh_y=3$.	06	CO2	BT4

=== ALL THE BEST ===

Bloom's Taxonomy (BT)

BT1 - Remember, BT2 - Understand, BT3 - Apply, BT4 - Analyze, BT5 - Evaluate, BT6 - Create

Course Outcomes:

	On successful completion of this course, students should be able to:
CO1	Learn CAD hardware and software for variety of applications
CO2	Compare different alternative facilities in CAD software
CO3	Implement the NC part programming and its application
CO4	Understand the different concepts in manufacturing system and its application for industrial automation.
CO5	Describe the robot and its different industrial applications.



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Department of Production Engineering
In-Semester Examination / Evaluation I
Academic Year: 2025-26 (semester I)

Class: Final Year B.Tech.

Elective: Project Management

Course Code: PEC-PE407

Date: 18.08.2025

Time: 11:00 to 11:45

Max. Marks: 20

- Attempt any two questions
- Assume suitable data whenever necessary and state it clearly.
- Figures to right indicate full marks.

Q.N.		Marks	CO	BT																																
Q.1	Cash flows for two project options are given below. Compare these projects by NPV method assuming a required rate of return of 12% and anticipated inflation rate of 3% over the lives of the two projects.	10	CO1	BT3																																
	<table><tr><td colspan="2">Project ABC</td><td colspan="2">Project XYZ</td></tr><tr><td colspan="2">Initial Investment = \$250000</td><td colspan="2">Initial Investment = \$400000</td></tr><tr><td colspan="2">Revenue:</td><td colspan="2">Revenue:</td></tr><tr><td>Year 1</td><td>\$ 50000</td><td>Year 1</td><td>\$ 75000</td></tr><tr><td>Year 2</td><td>\$ 100000</td><td>Year 2</td><td>\$ 250000</td></tr><tr><td>Year 3</td><td>\$ 100000</td><td>Year 3</td><td>\$ 250000</td></tr><tr><td>Year 4</td><td>\$ 200000</td><td>Year 4</td><td>\$ 50000</td></tr><tr><td>Year 5</td><td>\$ 75000</td><td>Year 5</td><td>\$ 25000</td></tr></table>	Project ABC		Project XYZ		Initial Investment = \$250000		Initial Investment = \$400000		Revenue:		Revenue:		Year 1	\$ 50000	Year 1	\$ 75000	Year 2	\$ 100000	Year 2	\$ 250000	Year 3	\$ 100000	Year 3	\$ 250000	Year 4	\$ 200000	Year 4	\$ 50000	Year 5	\$ 75000	Year 5	\$ 25000			
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Q.2	A company is considering investing in a new project opportunity. Top management is concerned about how long it will take for this new project to pay back the initial investment of \$75,000. The project should generate inflows of \$40,000, \$25,000, \$35,000, \$35,000, and \$25,000 for the next five years. The firm's required rate of return is 14%. How long will it take to pay back the initial investment?	10	CO1	BT2																																
Q.3	An entrepreneur has been working as a supplier of automobile spare parts to a number of retailers for the last 10 years. He has contacts with major automobile manufacturers from whom he buys the spares at company prices after negotiating suitable price discounts. He has now accumulated enough wealth to contemplate setting up of a manufacturing unit for a suspension unit used in 40% of the cars in the local market. Currently, the market share has been captured by two manufacturers who manufacture this suspension unit. And still the suspension units are often in short supply. Conduct a SWOT analysis for the entrepreneur indicating how well prepared he is to undertake his decision to switch from his current role as a supplier to a potential manufacturer.	10	CO1	BT4																																

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Course Outcomes:

	On successful completion of this course, students should be able to:
CO 1	Explore the feasibility of a project by applying financial and environmental criteria.



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Department of Production Engineering
In-Semester Examination / Evaluation I
Academic Year: 2025-26 (semester I)

Class: Final Year B.Tech.

Course: Costing and Estimation

Course Code: PEC-PE406

Date: 18.08.2025

Time: 11.00 am to 11.45 am

Max. Marks: 20

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Q.N.

- Q.1 A manufacturing concern produces a certain product in batches of 1000. The direct material cost and direct labour cost per batch of products are Rs 25500, and Rs. 55000 respectively. If 75% of direct material cost and 55% of direct labour cost are charged to compensate factory and administration cost on costs respectively, determine prime cost, factory cost, and production cost of each product manufactured. Mar
ks 10 CO BT
CO1 BT3
- Q.2 In a home appliances manufacturing unit, four sets of microwaves are manufactured, viz. Samsung, Whirlpool, Panasonic, and Midea. From the data given below, prepare a statement giving the cost per unit of each model in rupees and profit on each set sold. 10 CO2 BT4

	Samsung	Whirlpool	Panasonic	Midea
Direct Material	1025000	1275000	1349000	1125000
Direct Labour	175000	178326	148900	167125
Factory Overhead	70% of direct material	60% of direct labour	66% of direct material	30% of direct material & Labour
Office on-cost	40% of direct labour	40% of direct material	30% of direct material	15% of direct material & Labour
Units sold	989	950	899	998
Selling price	7999	8999	7499	6999

Estimate the cost per unit, profit earned per unit, and total profit earned by the factory.

=== ALL THE BEST ===

Bloom's Taxonomy (BT)

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Course Outcomes:

- On successful completion of this course, students should be able to:
- CO 1 Learn the basic concepts of cost, estimation, and depreciation fund calculation.
 - CO 2 Compute costs for various manufacturing processes like forging, welding, foundry etc.
 - CO 3 Understand the process of job costing, activity-based costing, cost accounting and budgetary control.
 - CO 4 Interpret during decision making, the concepts of CVP analysis, cost control techniques including time value of money.
 - CO 5 Implement the basics of engineering economics and financial management for profit making by the organization.



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SGGS INSTITUTE OF ENGINEERING & TECHNOLOGY, VISHNUPURI, NANDED – 431 606

Department of Production Engineering

Academic Year 2025-2026

ISE-1 Examination

Class: B.Tech. (Prod)

Subject: Productivity improvement techniques (PCC-PE-402)

Max Marks: 20

*CO: Course Outcomes.

NOTE: SOLVE ANY TWO QUESTIONS FROM THE FOLLOWING.

Q.1	Explain in detail the procedure for drawing an outline/operator process chart, and draw the same by taking any assembly product (having 3-4 components) as an example.	10	CO5	BT3
Q.2	Explain basic work content, excess work content and ineffective time. Also explain how the work content of a product/process increases? And suggest remedies to reduce the same.	10	CO1	BT1
Q.3	Explain in detail the examine phase in method study. Also explain what are the primary questions and secondary questions.	10	CO2	BT2